

Brief Insights

1. Customer Sales Performance

The analysis showed that a small number of customers contribute a significant portion of total sales. These high-value customers are important for revenue generation and should be targeted with loyalty programs and personalized offers.

2. Above-Average Sales Customers

Several customers generated sales higher than the overall average customer sales. These customers represent the most profitable segment and can be prioritized for retention and cross-selling opportunities.

3. Order-Level Analysis

Using subqueries, orders with sales greater than the average order value were identified. This helps the business understand which transactions contribute the most to revenue and highlights premium purchases.

4. Customer Ranking

Window functions were used to rank customers based on total sales. The ranking provides a clear view of the best-performing customers and helps management focus on the most valuable accounts.

5. Single-Order Customers

The analysis identified customers who placed only one order. These customers may require engagement strategies, promotions, or follow-up campaigns to encourage repeat purchases.

6. Highest Order Value per Customer

For each customer, the highest-value order was determined. This information helps understand customer purchasing behavior and identifies customers with the potential for large future transactions.

7. Data Analysis Techniques Used

The project successfully demonstrated the use of:

- **Subqueries** for filtering and comparison-based analysis.
- **Common Table Expressions (CTEs)** for modular and readable aggregation queries.
- **Window Functions (RANK and ROW_NUMBER)** for ranking and partition-based analysis.

- **JOIN operations** for combining customer and sales information into meaningful business reports.

Conclusion

By applying Subqueries, CTEs, Window Functions, and JOINS on the Superstore dataset, valuable insights into customer purchasing patterns and sales performance were obtained. The analysis helps identify top customers, low-performing customers, high-value orders, and potential opportunities for customer retention, enabling better business decision-making.